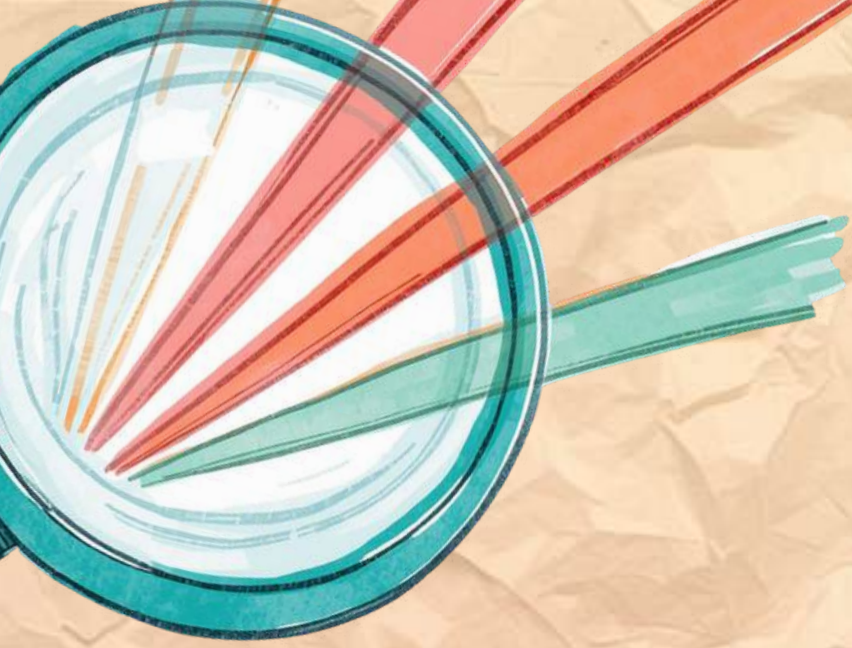




How They Bend Light and Make Things Bigger!

THE MAGIC OF LENSES



Grade 4
Optics
Academy





WHY IS ONE BLURRY AND THE OTHER
SO CLEAR?



Do you wear chashmey (glasses)?

YES

NO



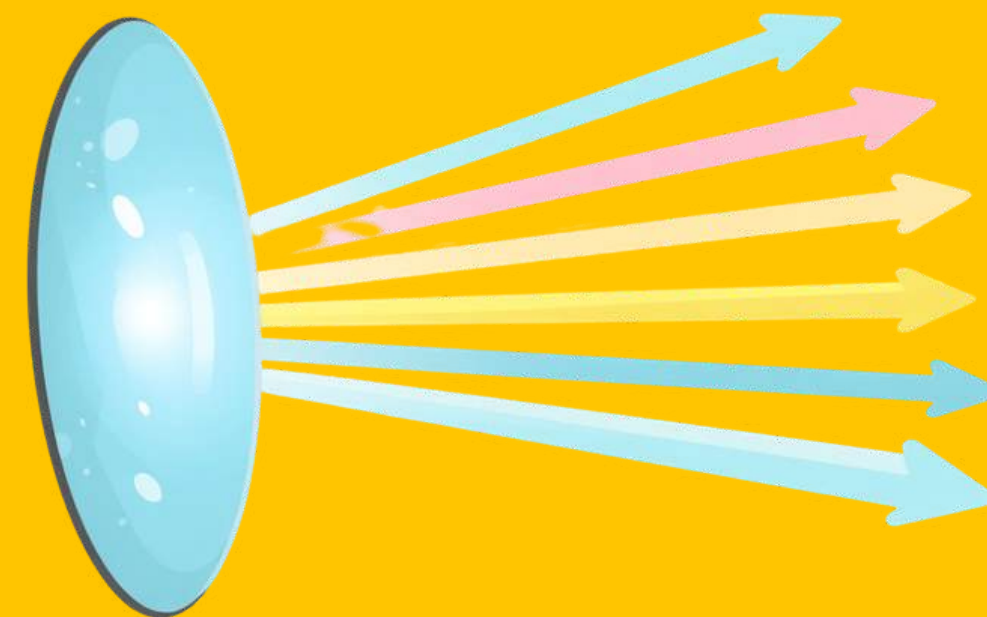
WHAT IS A LENS?

LENS

A lens is a clear, curved piece that bends light.
It helps us see things bigger or smaller!

Key Words to Remember:

- Lens – a curved piece of glass or plastic
- Transparent – lets light pass through
- Refraction – when light bends



REFRACTION IN LEMON WATER

BEND!

Light bends when it passes through different things like water or glass.
This is called refraction!
Look at the straw — it looks broken, but it's just light playing tricks!



CONVEX LENS THE MAGNIFIER

- 1 Convex lens is like a laddoo shape — thicker in the middle.
- 2 It bends light rays to come together (converge). It makes things BIGGER!
- 3 Memory trick: "Convex Comes OUT — Convex Converges — Convex = BIG!"



Remember!

laddoo = convex
thick in middle!



CONVEX LENS: DADI'S READING
CHASHMEY



MAGNIFYING GLASS MAGIC!

CONVEX LENS USES

SCHOOL MICROSCOPE

MOBILE CAMERA LENS

SCIENTISTS AND
PHOTOGRAPHERS
USE CONVEX LENSES TO
SEE BETTER!



CONCAVE LENS THE SHRINKER

- 1 Concave lens is like a cooking bowl — thinner in the middle.
- 2 It bends light rays apart (diverges). It makes things SMALLER!
- 3 Memory trick: "Concave CAVES IN like a bowl - Concave = SMALLER!"





MYOPIA GLASSES





DOOR PEEPHOLE (MAGIC EYE)

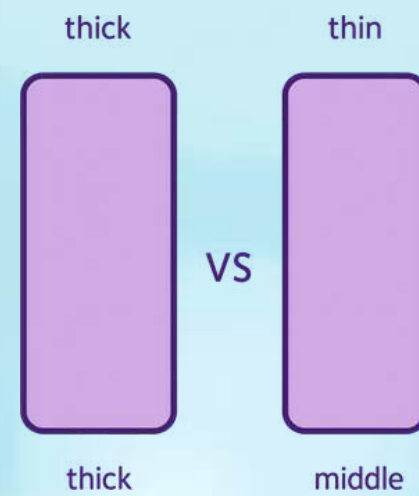




WIDE-ANGLE LENS

CONVEX VS CONCAVE: PART 1

CONVEX

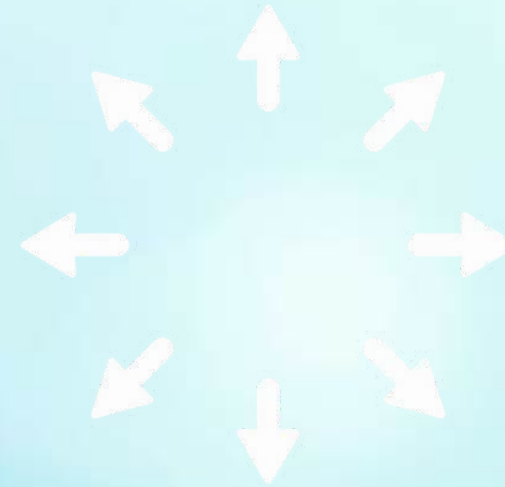


THICK IN MIDDLE

LIGHT RAYS CONVERGE



CONCAVE



RAYs DIVERGE

LENS USES COMPARISON

READING GLASSES



MICROSCOPE CONVEX



MYOPIA GLASSES



DOOR PEEPHOLE
CONCAVE



WIDE-ANGLE LENS



(Let's Find the Lenses!)

LENS DETECTIVE ACTIVITY!



🍷 LENS DETECTIVE 🍷

CAN YOU IDENTIFY WHICH
LENS TYPE EACH OBJECT
HAS?



DADI'S READING GLASSES → CONVEX

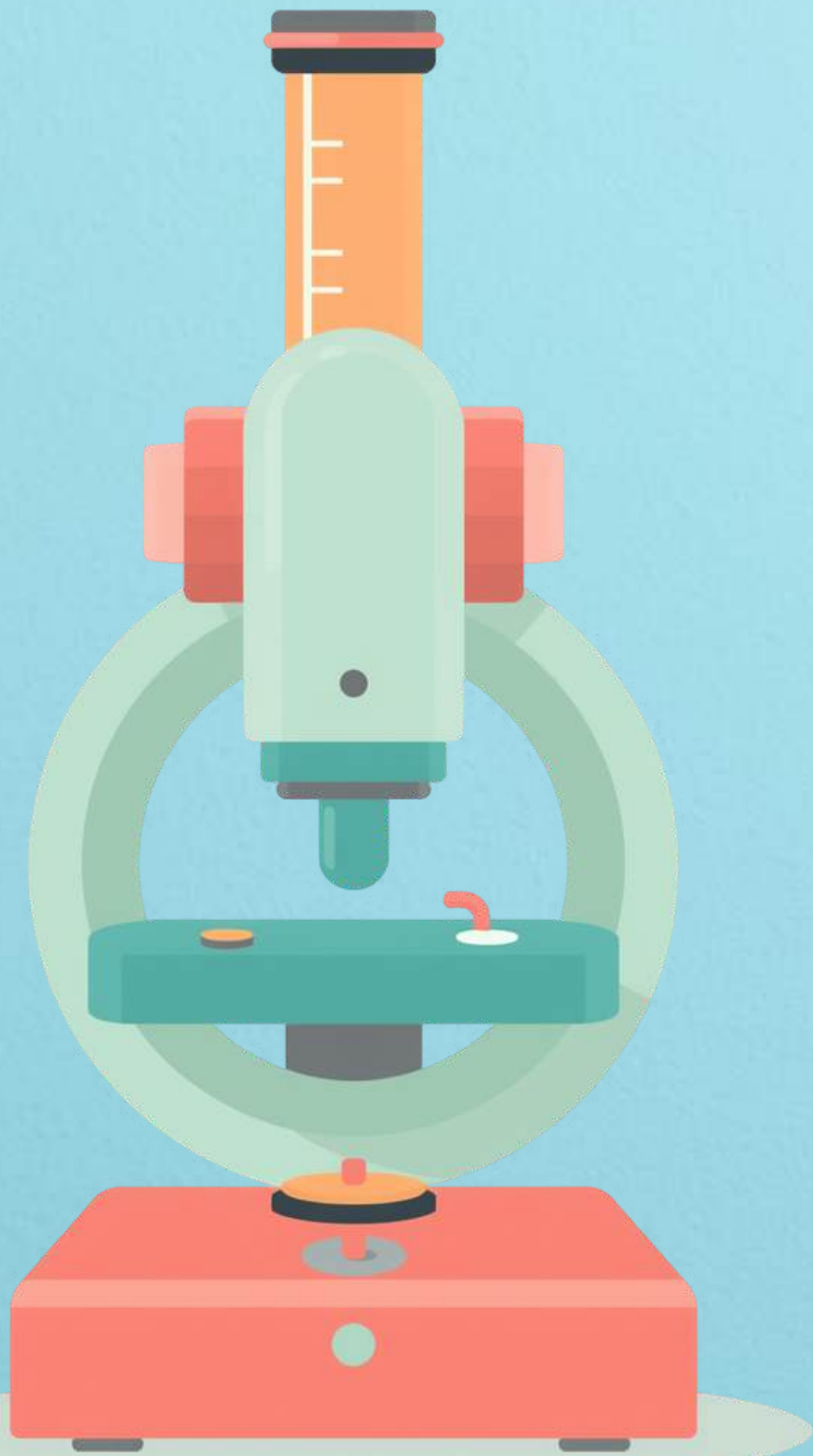
STUDENT'S MYOPIA GLASSES → CONCAVE

MAGNIFYING GLASS → CONVEX

WINDOW GLASS → NO LENS

🍌 LENS DETECTIVE PART 2 🍌

CAN YOU IDENTIFY WHICH LENS TYPE EACH OBJECT HAS?



DOOR PEEPHOLE → CONCAVE

MOBILE CAMERA → CONVEX

BATHROOM MIRROR → NOT A LENS!

SCHOOL MICROSCOPE → CONVEX

RECAP: WHAT DID WE LEARN?



Lens bends light — this is called refraction!



Convex = laddoo shape, converges light, makes things BIGGER!



Concave = Karahi shape, diverges light, makes things SMALLER!



Great Job!

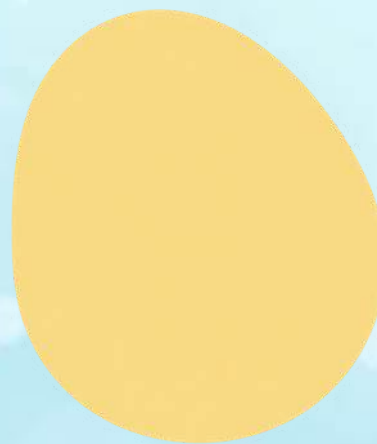
Indian examples help us see and explore!

VOCABULARY FLASHCARDS

LENS



REFRACTION



TRANSPARENT



CONVEX



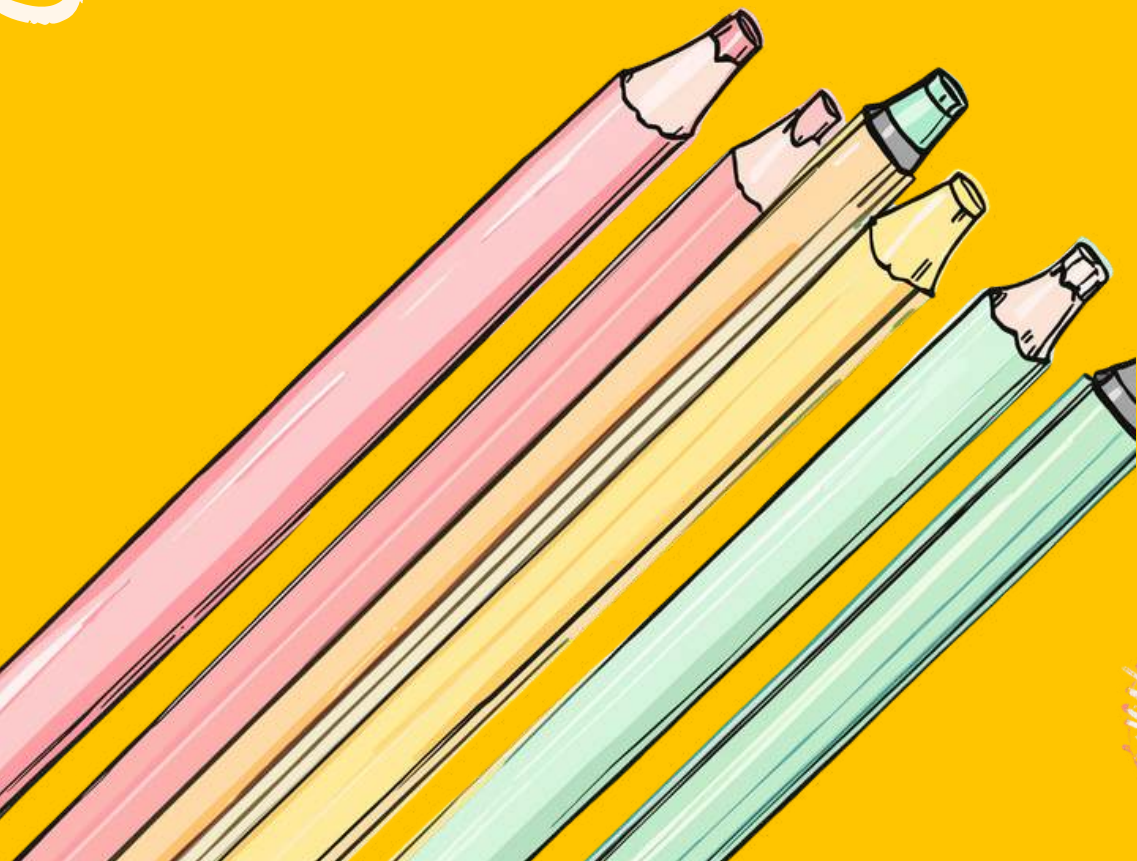
CONCAVE



LENS HUNT HOMEWORK



Find 3 objects with lenses at home.
Draw them in your notebook.
Write if each lens is convex or concave.
Ask your family to help you search!
Be creative with your drawings!





Thank You for Learning with Us!

**YOU'RE AMAZING
LENS DETECTIVES!**